



Cardiac Arrest: pVTach & VFib

Assessment

Pediatric Pearls:

- Use pediatric dosing of medications or electrical therapy for a pediatric patient < 37 kg and as defined by the PEDIA Tape.
- Focus on rapid and early BLS airway and ventilation tools. Intubation may not be the best option for these patients.
- Pediatric pads should be used in children < 10 Kg or PEDIA tape color purple.

Signs & Symptoms:

- Unresponsive
- Abnormal breathing (gasps)
- Pulseless
- Absent heart sounds
- Obvious death

Differential:

- Respiratory failure
- Foreign body airway obstruction
- Hyperkalemia
- Infection (Croup, epiglottitis)
- Hypovolemia
- Congenital heart disease
- Trauma
- Tension pneumothorax
- Hypothermia
- Toxins or Overdose
- Hypoglycemia
- Acidosis
- Acute MI or PE

Clinical Management Options

P	P	P	P	P	P
L	L	L	L	L	L
1	2	3	4	5	6

- Assess for unresponsiveness, absence of normal breathing, and pulselessness
- Assess for obvious death criteria
- [Begin Pit Crew CPR](#) procedure
- BLS [Airway Management](#) and [BVM](#) with Oxygen as available
- Passive oxygenation with nasal cannula at 25 LPM

- [Airway management](#) with [iGel](#) as needed
- Monitor [ETCO₂](#)

- [Vascular access](#) IO can be placed first, but IV access is preferred when able to obtain. If patient ≤ 12 months, distal femur IO is preferred
- [Epinephrine](#)
- Fluid bolus with [isotonic crystalloid](#) as needed (max 1000mL unless suspect hypovolemia)

- Monitor and interpret ECG

- [Manual Defibrillation](#)

- Adult: Maximum Joules

- Pediatric: Initial 2 joules/kg then repeat 4 joules/kg – Refer to Pediatric Dosing Chart or HandTevy

- [Amiodarone](#)

- If pVT/VF refractory to Amiodarone, then [Lidocaine](#)

- If Torsades de Points or polymorphic pVT, then [Magnesium Sulfate](#)

- If treatable cause is identified, move that that treatment up in priority

- Advanced Airway Management as needed. iGel is a good temporary and initial advanced airway, however, intubation should be performed when clinical and operational opportunities arise during the resuscitation.

- [Double sequential external defibrillation](#) at maximum Joules for Adults Only

- **IF** refractory to at least 2 shocks pads placed Anterior / Posterior, the 3rd shock should be DSED or vector change if 2nd monitor is not immediately available

- pVT / VF **NEVER** converted

- If [ROSC](#) then declare a [resuscitation alert](#) and use [Post Resuscitation Checklist](#).

Consult Online Medical Control As Needed



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PEDIATRIC DEFIBRILLATION DOSE CHART

Age	Preemie	Newborn	4MO	6MO	1YR	2YR	3YR	4YR	5YR	6YR	7YR	8YR	9YR	10YR	11YR	12YR	13YR
Weight	2KG	4KG	6KG	8KG	10KG	12KG	15KG	17KG	20KG	22KG	25KG	27KG	30KG	35KG	40KG	50KG	60KG
DEFIB #1	4j	8j	10j	15j	20j	20j	30j	30j	50j	50j	50j	50j	70j	70j	85j	100j	120j
DEFIB #2	8j	15j	20j	30j	50j	50j	70j	70j	85j	85j	100j	100j	120j	150j	150j	200j	200j

Pearls:

- Providers should not hasten departure after obtaining ROSC. Instead, focus on stabilizing the patient and ensuring that the airway is appropriately managed. If rearrest occurs, it will be in the first few minutes.
- Refer to drug formulary charts for all medication dosing for both adults and pediatric patients.
- In order to be successful in adult or pediatric arrests, a cause must be identified and corrected.
- Respiratory arrest is a common cause of pediatric cardiac arrest. Unlike adults, early oxygenation and ventilation is critical.
- In most cases pediatric airways can be managed by basic interventions.
- Effective CPR is critical: 1) Push hard and fast at appropriate rate 2) Ensure full chest recoil 3) Minimize interruptions in CPR. Pause CPR < 10 seconds only.
- Effective CPR and prompt defibrillation are the keys to successful resuscitation.
- Prolonged cardiac arrests may lead to tired providers and decreased compression quality. Ensure compressor rotation, summon additional resources as needed, and ensure provider rest and rehab during and post-event.
- For pediatrics use volume control device (IV Burette) for Dextrose and Fluid infusions
- Always quickly confirm asystole in more than one lead.
- Troubleshoot for equipment settings/problems
- PL1, PL2 and PL3 may only use automated external defibrillation (AED).
- Reassess and document ETT/BIAD placement continuously after every move and at transfer of patient care.
- Continuous ETCO₂ should be initiated as soon as possible.
- Calcium and sodium bicarbonate should be given early if hyperkalemia is suspected (renal failure, dialysis)
- Adult treatment priorities: uninterrupted compressions, defibrillation, ventilation, then IV/IO and airway management if needed.
- Polymorphic VT (Torsades) may benefit from Magnesium Sulfate.
- Prior to any external shocks providers should verify that defibrillation pads are well-adhered to the patient and that they do not touch.
- Continue to use **primary monitor** for all event recording and data capture.
- All monitor event data and recordings are uploaded into e-PCR.
- Once criteria for DSED are met subsequent shocks should be delivered as DSED
- Ultrasound to determine cardiac wall motion at pulse check for patients in PEA rhythms (NOT asystole); DO NOT interrupt compressions for ultrasound.
- Ensure that 2 cardiac monitors are brought into the patient's side to ensure that DSED can be performed in a timely manner, if indicated.